

QUIZ-4

Time: 20 minutes

Marks: 15

Instructions: Answer all questions. Show necessary calculations and reasoning.

Question 1

[5 Marks]

A word embedding model is trained using:

MovieA is great, MovieB is great

Vocabulary order:

[MovieA, is, great, MovieB]

Embedding matrix:

$$E = \begin{bmatrix} 0.38 & 0.42 \\ 0.10 & 0.90 \\ 1.85 & 0.98 \\ 0.36 & 0.40 \end{bmatrix}$$

For input word MovieA, output probabilities are:

$$P(\text{MovieA}) = 0.23, \quad P(\text{is}) = 0.27, \quad P(\text{great}) = 0.26, \quad P(\text{MovieB}) = 0.24$$

The correct next word is is.

- (a) Represent MovieA using one-hot encoding. [1]
- (b) Find the embedding vector of MovieA. [1]
- (c) Which word will the model predict? [1]
- (d) Compute the cross-entropy loss for this prediction. [1]
- (e) Explain why the embeddings of MovieA and MovieB may become close after training. [1]

Question 2

[10 Marks]

A Transformer encoder receives:

$$X_L = [1.87, 1.09], \quad X_G = [-1.68, 0.67]$$

where L = Lets and G = go. The encoder matrices are:

$$W_Q = \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}, \quad W_K = \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix}, \quad W_V = \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix}$$

- (a) Compute Q_L, K_L, V_L for "Lets". [3]
- (b) Compute Q_G, K_G, V_G for "go". [3]
- (c) Using rounded values

$$Q_G = [-3.7, 0.6], \quad K_L = [-4.7, 1.9], \quad K_G = [1.9, -0.2],$$

compute the attention scores of "go" with "Lets" and "go". [2]

- (d) If the softmax weights for "go" are $[1.0, 0.0]$, and

$$V_L = [2.5, -2.1], \quad V_G = [-2.7, 1.5],$$

compute the attention output for "go". [2]